Tennessee TECH

Quiz Questions

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Data parallelism _____.

- divides the program's logic into independent tasks that can be executed on different processors simultaneously (in parallel).
- divides data into chunks that are processed simultaneously (in parallel) using the same operation.
- is a form of parallel execution where different tasks are performed concurrently on multiple processors.
- uses multiple processors to each run the entire dataset independently to compare results.

Task parallelism _____.

- uses multiple processors to each run the entire dataset independently to compare results.
- is the process of distributing work across processors to ensure that no processor is significantly more burdened than others.
- is a form of parallel execution where different tasks are performed concurrently on multiple processors.
- divides data into chunks that are processed simultaneously (in parallel) using the same operation.

Which sorting algorithm is best suited for parallelization due to its divide-and-conquer nature?

- Bubble sort
- Selection sort
- Merge sort
- Insertion sort
- Radix sort

Which type of parallelism involves applying the same task to different subsets of data?

- Task Parallelism
- Loop Unrolling
- Data Parallelism
- Thread Pooling
- Pipelining
- Load Balancing

What happens if you select more threads than the number of processor cores available on your system?

- It increases efficiency and therefore runs faster.
- It has no impact and only uses the number of threads available on your system.
- It will likely slow down the execution time due to context switching (switching between processors).
- It will cause a compilation error.

Which of the following best describes a thread in parallel programming?

- A separate program running in its own memory space.
- A variable that stores shared data.
- A mini-program that shares memory and executes part of a task concurrently.
- An error handler in C++ and other languages.

What is the primary goal of using OpenMP in C++ programs?

- The goal is to reduce the number of lines of code.
- The goal is to increase memory usage.
- The goal is to speed up programs using multiple CPU cores (processors).
- The goal is to prevent recursion overhead in algorithms.

What header file must you include to use the OpenMP library?

- opmp.h
- openm_parallel.h
- openmp.h
- omp.h

Which OpenMP directive is used to define a parallel region?

- omp_set_num_threads()
- #pragma omp parallel
- #pragma omp critical
- #pragma omp

Which OpenMP function returns the unique ID of a thread?

- omp_get_thread_num()
- omp_id()
- omp_get_thread_count()
- omp_thread_id()

True or False. Each thread in OpenMP has its own separate memory space.

- Always true because each thread can only see the variables it is working with
- Always false because variables are shared among threads and can't be declared private
- False, but variables can be declared private so that each thread gets its own copy of the variable
- None of the above